

Project Proposal

Goal: This project aims to build an emotion-aware music recommendation system that goes beyond traditional acoustic or co-listening signals. Instead of collaborative filtering, we use lyric's emotional semantics to better align recommendations with intended vibe.

Objective: We will build a dynamic hybrid recommendation system that combines playlist-based collaborative filtering with lyric-based emotion analysis. The system first learns long-term playlist preference from song co-occurrence patterns, then dynamically adjusts recommendation weights based on recent emotional shifts within the playlist.

Problem: Current commercial systems (Spotify/Apple Music) primarily rely on Collaborative Filtering (user-item interaction) and basic audio features. These systems suffer from:

- **Semantic Blindness:** They cannot distinguish between a "sad" song about heartbreak and a "sad" song about grief.
- **Cold-Start Sensitivity:** They struggle to recommend new or niche songs that lack massive play history but possess the correct "emotional vibe."

Our Solution: We combine collaborative filtering with LLM-based lyric emotion analysis, dynamically weighting long-term playlist preference and recent mood signals to recommend the next songs.

Novelty Justification:

1. Algorithmic Novelty: *Enhanced Emotion-aware Music Recommendation via Large Language Models (LEMON)* — **ACM KDD 2025**.

Role: Provides the framework for hierarchical emotional extraction from lyrics.

2. Architectural Novelty: *Towards Open-World Recommendation with Knowledge Augmentation from LLMs (KAR)* — **ACM RecSys 2023**.

Role: Justifies using LLMs to provide cultural context that traditional datasets lack.

3. Lyric/Summary-Based Semantic Features: *Music Recommendation through LLM Song Summary* — **ROEGEN@RecSys 2024**.

Role: Supports using LLM-generated textual representations for music recommendation.

Dataset:

- **Spotify Million Playlist Dataset (MPD):** Contains playlist titles, track lists, artist/album/track metadata, and will be used as the main user-preference dataset, where each playlist is treated as a proxy user profile.
- **Spotify Million Song Dataset:** Contains song titles, artist names, links, and lyrics, and will be used to match MPD tracks and extract lyric-based emotional/semantic features.

Evaluation Metric:

- We will use the first 80% of each playlist as the input and hold out the last 20% as unseen test songs. The model will recommend new songs based on the input playlist, and we will evaluate whether the recommendations match the held-out 20%. This allows us to test whether recent short-term mood signals improve prediction beyond the playlist's long-term preference.
- We will measure performance using Recall@10 and NDCG@10, which evaluate whether the held-out songs appear in the top-10 recommendations and whether they are ranked near the top.

Proposed Approach:

- First, generate a simple user preference using the Spotify Million Playlist Dataset to learn playlist-level song co-occurrence patterns.
- Second, we will use a low-parameter open-source LLM (such as llama3 or gemini 1.5) to assign emotional tags or emotion embeddings to each song. The LLM does better in interpreting metaphor, indirect emotional expression, and multilingual lyrics.
- Third, we analyze the emotional pattern of the observed 80% of each playlist. We attempt to detect emotional divergence points, where the playlist shifts from one emotional style to another. Such as begins from calm songs but shifts toward sad or energetic tracks. We estimate the average emotional switching point or switching frequency within the observed playlist.
- Finally we combine both long-term preference and short time mood, create a better prediction of the next song in the playlist.

Limitation:

- Since no access to real individual user profiles and listening histories. Therefore, we treat playlists as proxy users. A playlist may be created for collection, sharing, archiving, and temporary events rather than actual listening behavior or emission state.
- MPD reflects playlist behavior from 2010–2017, while the lyrics dataset does not provide a clearly documented collection period. Therefore, the combined dataset may not represent current music trends, recent artists, or modern Spotify usage patterns.

Ethics:

- Risk of emotional spiraling. If a user provides a sad or distressed playlist context, an emotion-aware recommender may repeatedly recommend more negative songs, potentially deepening the emotional state. The system should include diversity constraints and avoid recommending only one emotional category for long sequences.
- Lyric interpretation can contain cultural, linguistic, and genre bias. LLMs may misinterpret irony, metaphor, slang, multilingual lyrics, or culturally specific expressions.

Citation:

<https://www.kaggle.com/datasets/himanshuwagh/spotify-million>

2017 - 2020 Spotify playlist

<https://www.kaggle.com/datasets/notshrirang/spotify-million-song-dataset>

Spotify millions song data

<https://dblp.org/rec/conf/kdd/WangOZXRP25>

KDD 2025. Uses LLMs for emotion-aware music recommendation by modeling song emotions, short-term user mood, and long-term preferences.

https://roegen-recsys2024.github.io/papers/recsys2024-workshops_paper_208.pdf

ROEGEN@RecSys 2024. Uses LLM-generated song summaries as textual features for a recommendation model. Very practical for a course project.

<https://arxiv.org/abs/2306.10933>

RecSys 2024. Not purely a music recommendation paper, but it applies LLM-based knowledge augmentation to recommender systems and reports deployment on music recommendation platforms.

